

Pricing Bond Options Using the Forward Measure vs. Risk-Neutral Measure

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This document explains how to price a bond option using two different probability measures: the standard risk-neutral measure and the T-forward measure.

1 Method 1: Using the T-Forward Measure (Standard Practice)

The forward measure simplifies the pricing process by changing the numéraire (the unit of account) from cash to the zero-coupon bond that matures at the option's exercise date.

1.1 General Methodology

1. **Choose the Numéraire:** Select the zero-coupon bond (ZCB) maturing at the option's exercise date, T_{option} , with price $P(t, T_{\text{option}})$ at time t .
2. **Change of Measure:** Apply Girsanov's theorem to move from the risk-neutral measure to the T_{option} -forward measure ($\mathbb{Q}_{T_{\text{option}}}$). Under this measure, asset prices expressed in terms of the numéraire are martingales.
3. **Price the Option:** The price of the option today, C_0 , is the discounted expected value of its payoff, where the expectation is taken under the forward measure.

1.2 Pricing a European Call Option Formula

For a European call option with exercise date T_{option} and strike price K , on a zero-coupon bond maturing at T_{bond} ($T_{\text{bond}} > T_{\text{option}}$), the price at time t is:

$$C_t = P(t, T_{\text{option}}) \times E^{\mathbb{Q}_{T_{\text{option}}}}[\max(P(T_{\text{option}}, T_{\text{bond}}) - K, 0) | \mathcal{F}_t]$$

1.2.1 Using the Black Model (Closed-Form Solution)

Assuming the forward bond price is log-normally distributed, we use the Black model formula:

$$C_t = P(t, T_{\text{option}})[F_t N(d_1) - KN(d_2)]$$

where:

- F_t is the forward price of the underlying bond at time t .
- K is the strike price.
- $N(\cdot)$ is the cumulative standard normal distribution function.
- $d_1 = \frac{\ln(F_t/K) + (\sigma^2/2)T_{\text{exp}}}{\sigma\sqrt{T_{\text{exp}}}}$
- $d_2 = d_1 - \sigma\sqrt{T_{\text{exp}}}$
- T_{exp} is the time to option expiration ($T_{\text{option}} - t$).
- σ is the volatility of the forward bond price.

2 Method 2: Using the Risk-Neutral Measure (More Complex)

It is theoretically possible to price the option using the risk-neutral measure ($\mathbb{Q}$), but it is significantly more complex due to the stochastic nature of interest rates.

2.1 The Formula Under the Risk-Neutral Measure

The price is given by the fundamental risk-neutral pricing formula:

$$C_t = E^{\mathbb{Q}}[e^{-\int_t^{T_{\text{option}}} r_s ds} \times \max(P(T_{\text{option}}, T_{\text{bond}}) - K, 0) | \mathcal{F}_t]$$

2.2 Challenges

The primary difficulty lies in dealing with the expectation of a product of two **correlated stochastic variables**:

1. **The Option Payoff:** The underlying bond price $P(T_{\text{option}}, T_{\text{bond}})$ at expiration.
2. **The Stochastic Discount Factor:** The interest rates r_s between t and T_{option} .

To solve this under the risk-neutral measure, one must:

- Specify a complex interest rate model (e.g., HJM, LMM) that describes the entire term structure.
- Solve a mathematically challenging problem involving the joint probability distribution of the underlying bond price and the stochastic discount factor.

3 Conclusion

The T-forward measure is the standard industry practice because it simplifies the calculation by removing the stochastic discount factor from the expectation, leading to tractable closed-form solutions like the Black model.